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Adomian decomposition method for a reliable treatment of the Bratu-type equations

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Abstract

In this paper, we present a framework to determine exact solutions of Bratu-type equations. The algorithm rests mainly on the Adomian decomposition method. The proposed scheme is illustrated by studying two boundary value problems and an initial value problem of Bratu-type. The first type gives a solution that blows up at the middle of the domain, whereas the other equations give bounded solutions.
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1. Introduction

This paper is concerned with boundary value problems and an initial value problem of the Bratu-type. It is well known that Bratu's boundary value problem in one-dimensional planar coordinates is of the form [1–6]

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$$\begin{aligned} &u'' + \lambda e^u = 0, \quad 0 < x < 1, \\ &u(0) = u(1) = 0. \end{aligned} \tag{1}$$

The standard Bratu problem (1) was used to model a combustion problem in a numerical slab. The Bratu model appears in a number of applications such as the fuel ignition of the thermal combustion theory [4] and in the Chandrasekhar model of the expansion of the universe. It stimulates a thermal reaction process in a rigid material where the process depends on the balance between chemically generated heat and heat transfer by conduction [6].

A substantial amount of research work has been directed for the study of the Bratu problem [1–6]. Several numerical techniques, such as the finite difference method, finite element approximation, weighted residual method [6], and the shooting method [1], have been implemented independently to handle the Bratu model numerically. In addition, Boyd [2,3] employed Chebyshev polynomial expansions and the Gegenbauer polynomials as base functions.

The exact solution to (1) is given in [1] and [2,3] and given by

$$u(x) = -2\ln\left[\frac{\cosh\left(\left(x-\frac{1}{2}\right)\frac{\theta}{2}\right)}{\cosh\left(\frac{\theta}{4}\right)}\right], \tag{2}$$

where θ satisfies

$$\theta = \sqrt{2\lambda}\cosh\left(\frac{\theta}{4}\right). \tag{3}$$

The Bratu problem has zero, one or two solutions when $\lambda > \lambda_c$, $\lambda = \lambda_c$, and $\lambda < \lambda_c$, respectively, where the critical value λ_c satisfies the equation

$$1 = \frac{1}{4}\sqrt{2\lambda_c}\sinh\left(\frac{\theta_c}{4}\right). \tag{4}$$

It was evaluated in [1–6] that the critical value λ_c is given by

$$\lambda_c = 3.513830719. \tag{5}$$

The main goal of this problem is to introduce a reliable treatment of two boundary value problems of Bratu-type model, given by

$$\begin{aligned} &u'' - \pi^2 e^u = 0, \quad 0 < x < 1, \\ &u(0) = u(1) = 0, \end{aligned} \tag{6}$$

and

$$\begin{aligned} &u'' + \pi^2 e^{-u} = 0, \quad 0 < x < 1, \\ &u(0) = u(1) = 0. \end{aligned} \tag{7}$$

In addition, an initial value problem of the Bratu-type

$$\begin{aligned} &u'' - 2e^u = 0, \ 0 < x < 1, \\ &u(0) = u'(0) = 0, \end{aligned} \tag{8}$$

will be investigated.

In this paper, our work stems mainly from the Adomian decomposition method [7–14]. The Adomian decomposition method, which accurately computes the series solution, is of great interest to applied sciences. The method provides the solution in a rapidly convergent series with components that are elegantly computed.

The main advantage of the method is that it can be applied directly for all types of differential and integral equations, linear or nonlinear, homogeneous or inhomogeneous, with constant coefficients or with variable coefficients. Another important advantage is that the method is capable of greatly reducing the size of computational work while still maintaining high accuracy of the numerical solution. Convergence of Adomian decomposition scheme was established by many authors by using fixed point theorems. The effectiveness and the usefulness of Adomian method are demonstrated by finding exact solutions to the models that will be investigated. For concrete problems, the method provides approximations that provide high accuracy by using few iterations.

2. Analysis of the method

Adomian decomposition method [7–14] defines the unknown function $u(x)$ by an infinite series

$$u(x) = \sum_{n=0}^{\infty} u_n(x), \tag{9}$$

where the components $u_n(x)$ are usually determined recurrently. The nonlinear operator $F(u)$ can be decomposed into an infinite series of polynomials given by

$$F(u) = \sum_{n=0}^{\infty} A_n, \tag{10}$$

where A_n are the so-called Adomian polynomials of $u_0, u_1, \ldots, u_n$ defined by

$$A_n = \frac{1}{n!} \frac{d^n}{d\lambda^n} \left[F\left(\sum_{i=0}^{n} \lambda^i \, u_i \right) \right]_{\lambda=0}, \quad n = 0, 1, 2, \ldots, \tag{11}$$

or equivalently

$$
\begin{aligned}
&A_0 = F(u_0),\\
&A_1 = u_1F'(u_0),\\
&A_2 = u_2F'(u_0) + \frac{1}{2}u_1^2F''(u_0),\\
&A_3 = u_3F'(u_0) + u_1u_2F''(u_0) + \frac{1}{3}u_1^3F'''(u_0),\\
&A_4 = u_4F'(u_0) + \left(u_1u_3 + \frac{1}{2}u_2^2\right)F''(u_0) + \frac{1}{2}u_1^2u_2F'''(u_0) + \frac{1}{24}u_1^4F^{(iv)}(u_0),\\
&\ldots
\end{aligned}
\tag{12}
$$

It is now well known that these polynomials can be generated for all classes of nonlinearity according to specific algorithms defined by (11). Recently, an alternative algorithm for constructing Adomian polynomials has been developed by Wazwaz [13].

3. First Bratu-type problem

We first consider the Bratu-type model

$$
\begin{aligned}
&u'' - \pi^2 e^u = 0, \quad 0 < x < 1,\\
&u(0) = u(1) = 0.
\end{aligned}
\tag{13}
$$

As indicated before, (13) differs from Bratu problem by the value of λ given by

$$\lambda = -\pi^2 < 0.$$

However, in (1), $\lambda = \lambda_c = 3.513830719 > 0$. The effect of this change in λ will be examined.

To begin with, (13) can be written in an operator form

$$
\begin{aligned}
&Lu = \pi^2 e^u,\\
&u(0) = u(1) = 0,
\end{aligned}
\tag{14}
$$

where the differential operator L is

$$L = \frac{\partial^2}{\partial x^2}. \tag{15}$$

The inverse L^{-1} is assumed to be a two-fold integral operator given by

$$L^{-1}(\cdot) = \int_0^x \int_0^x (\cdot)\, \mathrm{d}x\, \mathrm{d}x. \tag{16}$$

Applying the inverse operator L^{-1} on both sides of (14) and using the initial condition $u(0) = 0$ we find

$$u(x) = ax + L^{-1}(\pi^2 e^u), \tag{17}$$

where $a = u'(0) \neq 0$ is not given but will be determined by using the other boundary condition. Substituting (9) and (10) into the functional equation (14) gives

$$\sum_{n=0}^{\infty} u_n(x) = ax + L^{-1}\left(\pi^2 \sum_{n=0}^{\infty} A_n\right), \tag{18}$$

where A_n are the so-called Adomian polynomials. Identifying the zeroth component $u_0(x)$ by ax, the remaining components $u_n(x)$, $n \geqslant 1$ can be determined by using the recurrence relation

$$\begin{aligned} &u_0(x) = ax, \\ &u_{k+1}(x) = \pi^2 L^{-1}(A_k), \quad k \geqslant 0, \end{aligned} \tag{19}$$

where A_k are Adomian polynomials that represent the nonlinear term e^u and given by

$$\begin{aligned} A_0 &= e^{u_0}, \\ A_1 &= u_1 e^{u_0}, \\ A_2 &= \left(u_2 + \frac{1}{2}u_1^2\right)e^{u_0}, \\ A_3 &= \left(u_3 + u_1 u_2 + \frac{1}{6}u_1^3\right)e^{u_0}, \\ A_4 &= \left(u_4 + u_1 u_3 + \frac{1}{2}u_2^2 + \frac{1}{2}u_1^2 u_2 + \frac{1}{24}u_1^4\right)e^{u_0}, \\ &\ldots \end{aligned} \tag{20}$$

Other polynomials can be generated in a similar way to enhance the accuracy of approximation.

Combining (19) and (20) yields

$$\begin{aligned} u_0(x) &= ax, \\ u_1(x) &= -\frac{\pi^2}{a^2}(-e^{ax} + ax + 1), \\ u_2(x) &= -\frac{\pi^4}{4a^4}(-e^{2ax} + 4axe^{ax} - 4e^{ax} + 2ax + 5), \\ u_3(x) &= \frac{\pi^6}{12a^6}(e^{3ax} + 6e^{2ax}(1 - ax) + 3e^{ax}(2a^2x^2 - 6ax + 5) - 6ax - 22), \\ &\ldots . \end{aligned} \tag{21}$$

It is, in principle, possible to calculate more components in the decomposition series to enhance the approximation. In view of (21), the solution $u(x)$ is readily obtained in a series form by

$$u(x) = ax - \frac{\pi^2}{a^2}(-e^{ax} + ax + 1) - \frac{\pi^4}{4a^4}(-e^{2ax} + 4axe^{ax} - 4e^{ax} + 2ax + 5) + \frac{\pi^6}{12a^6}(e^{3ax} + 6e^{2ax}(1 - ax) + 3e^{ax}(2a^2x^2 - 6ax + 5) - 6ax - 22) + \cdots, \tag{22}$$

or equivalently

$$u(x) = ax + \frac{\pi^2}{2!}x^2 + \frac{\pi^2 a}{3!}x^3 + \left(\frac{\pi^2a^2 + \pi^4}{4!}\right)x^4 + \left(\frac{\pi^2a^3 + 4\pi^4 a}{5!}\right)x^5 + \left(\frac{11\pi^4a^2 + \pi^2a^4 + 4\pi^6}{6!}\right)x^6 + \left(\frac{26\pi^4a^3 + \pi^2a^5 + 34\pi^6 a}{6!}\right)x^7 + \cdots, \tag{23}$$

obtained upon using Taylor series. The exact solution can now be obtained by considering the boundary condition $u(1) = 0$ to obtain that $a = \pi$, and consequently the closed form solution

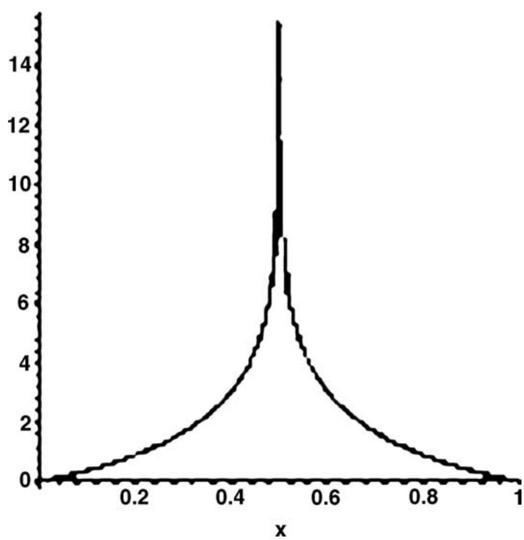

Fig. 1. The solution of the first Bratu-type equation where the solution blows up at $x = 0.5$ and bounded elsewhere.

$$u(x) = -\ln\left(1 + \cos\left(\left(\frac{1}{2} + x\right)\pi\right)\right). \tag{24}$$

It is interesting to point out that unlike the solution of the standard Bratu equation where $u(x) < \infty$ for all values of x of its domain, the solution (24) blows up at $x = 0.5$ as shown by Fig. 1.

4. Second Bratu-type problem

We next consider the Bratu-type model

$$\begin{aligned} &u'' + \pi^2 e^{-u} = 0,\ \ 0 < x < 1, \\ &u(0) = u(1) = 0. \end{aligned} \tag{25}$$

(25) differs from the standard Bratu problem by the term e^{-u}, and

$$\lambda = \pi^2 > \lambda_c.$$

The effect of these changes will be examined.

To begin with, (25) can be written in an operator form

$$\begin{aligned} &Lu = -\pi^2 e^{-u}, \\ &u(0) = u(1) = 0, \end{aligned} \tag{26}$$

where the differential operator L is

$$L = \frac{\partial^2}{\partial x^2}. \tag{27}$$

The inverse L^{-1} is assumed a two-fold integral operator given by

$$L^{-1}(\cdot) = \int_0^x \int_0^x (\cdot)\ \mathrm{d}x\ \mathrm{d}x. \tag{28}$$

Applying the inverse operator L^{-1} on both sides of (26) and using the initial condition $u(0) = 0$ we find

$$u(x) = ax - L^{-1}(\pi^2 e^{-u}), \tag{29}$$

where $a = u'(0) \neq 0$ is not given but will be determined by using the other boundary condition. Substituting (9) and (10) into the functional Eq. (26) gives

$$\sum_{n=0}^{\infty} u_n(x) = ax - L^{-1}\left(\pi^2 \sum_{n=0}^{\infty} B_n\right), \tag{30}$$

where B_n are the so-called Adomian polynomials of the nonlinear term e^{-u}. Identifying the zeroth component $u_0(x)$ by ax, the remaining components $u_n(x)$, $n \geqslant 1$ can be obtained recurrently by using the relation

$$\begin{aligned} &u_0(x) = ax, \\ &u_{k+1}(x) = -\pi^2 L^{-1}(B_k), \quad k \geqslant 0, \end{aligned} \tag{31}$$

where B_k are Adomian polynomials that represent the nonlinear term e^{-u} and given by

$$\begin{aligned} &B_0 = e^{-u_0}, \\ &B_1 = -u_1 e^{-u_0}, \\ &B_2 = \left(-u_2 + \frac{1}{2}u_1^2\right)e^{-u_0}, \\ &B_3 = \left(-u_3 + u_1 u_2 - \frac{1}{6}u_1^3\right)e^{-u_0}, \\ &B_4 = \left(-u_4 + u_1 u_3 + \frac{1}{2}u_2^2 - \frac{1}{2}u_1^2 u_2 + \frac{1}{24}u_1^4\right)e^{-u_0}, \\ &\cdots \end{aligned} \tag{32}$$

Other polynomials can be generated in a similar way to enhance the accuracy of approximation.

Combining (31) and (32) yields

$$\begin{aligned} &u_0(x) = ax, \\ &u_1(x) = -\frac{\pi^2}{a^2}(e^{-ax} + ax - 1), \\ &u_2(x) = -\frac{\pi^4}{4a^4}(e^{-2ax} + 4axe^{-ax} + 4e^{-ax} + 2ax - 5), \\ &u_3(x) = -\frac{\pi^6}{12a^6}(e^{-3ax} + 6e^{-2ax}(1 + ax) + 3e^{-ax}(2a^2x^2 + 6ax + 5) \\ &\qquad\quad + 6ax - 22), \\ &\cdots. \end{aligned} \tag{33}$$

More components in the decomposition series can be computed to enhance the approximation. In view of (33), the solution $u(x)$ is readily obtained in a series form by

$$\begin{aligned} u(x) = {} & ax - \frac{\pi^2}{a^2}(e^{-ax} + ax - 1) - \frac{\pi^4}{4a^4}(e^{-2ax} + 4axe^{-ax} + 4e^{-ax} + 2ax - 5) \\ & - \frac{\pi^6}{12a^6}(e^{-3ax} + 6e^{-2ax}(1 + ax) + 3e^{-ax}(2a^2x^2 + 6ax + 5) + 6ax - 22) \\ & + \cdots, \end{aligned} \tag{34}$$

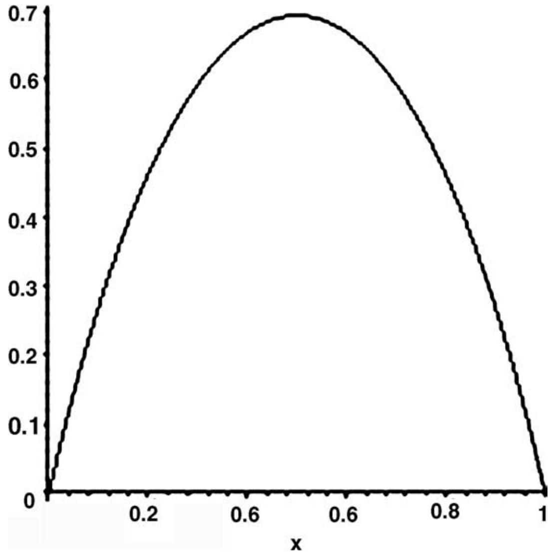

Fig. 2. The solution of the second Bratu-type where $u(x)$ < infty for $0 \leqslant x \leqslant 1$.

or equivalently

$$u(x) = ax - \frac{\pi^2}{2!}x^2 + \frac{\pi^2 a}{3!}x^3 - \left(\frac{\pi^2 a^2 + \pi^4}{4!}\right)x^4 + \left(\frac{\pi^2 a^3 + 4\pi^4 a}{5!}\right)x^5 - \left(\frac{11\pi^4 a^2 + \pi^2 a^4 + 4\pi^6}{6!}\right)x^6 + \left(\frac{26\pi^4 a^3 + \pi^2 a^5 + 34\pi^6 a}{6!}\right)x^7 + \cdots. \tag{35}$$

The exact solution can now be obtained by considering the boundary condition $u(1) = 0$ to obtain that $a = \pi$, and consequently the closed form solution

$$u(x) = \ln(1 + \sin(1 + \pi x)). \tag{36}$$

It is interesting to point out that the solution (36) is bounded where $u(x) < \infty$ for all values of its domain $0 \leqslant x \leqslant 1$ as shown by Fig. 2.

5. Initial value problem of the Bratu-type

We finally consider the initial value problem of Bratu-type

$$\begin{aligned} &u'' - 2e^u = 0, \quad 0 < x < 1, \\ &u(0) = u'(0) = 0. \end{aligned} \tag{37}$$

Unlike the Bratu problem where boundary conditions are used, (37) is an initial value problem where $u(0) = u'(0) = 0$.

To begin with, (37) can be written in an operator form

$$\begin{aligned} & Lu = 2e^u, \\ & u(0) = u(1) = 0, \end{aligned} \tag{38}$$

where the differential operator L is

$$L = \frac{\partial^2}{\partial x^2}. \tag{39}$$

The inverse L^{-1} is assumed a two-fold integral operator given by

$$L^{-1}(\cdot) = \int_0^x \int_0^x (\cdot)\, \mathrm{d}x\, \mathrm{d}x. \tag{40}$$

Applying the inverse operator L^{-1} on both sides of (38) and using the initial condition $u(0) = u'(0) = 0$ we find

$$u(x) = 2L^{-1}(e^u). \tag{41}$$

Substituting (9) and (10) into the functional Eq. (41) gives

$$\sum_{n=0}^{\infty} u_n(x) = 2L^{-1}\left(\sum_{n=0}^{\infty} A_n\right), \tag{42}$$

where A_n are the so-called Adomian polynomials of the nonlinear term e^u defined above in (20). Identifying the zeroth component $u_0(x)$ by 0, the remaining components $u_n(x)$, $n \geqslant 1$ can be obtained recurrently by using the relation

$$\begin{aligned} & u_0(x) = 0, \\ & u_{k+1}(x) = 2L^{-1}(A_k), \quad k \geqslant 0. \end{aligned} \tag{43}$$

This in turn gives

$$\begin{aligned} & u_0(x) = 0, \\ & u_1(x) = x^2, \\ & u_2(x) = \frac{1}{6}x^4, \\ & u_3(x) = \frac{2}{45}x^6, \\ & u_4(x) = \frac{17}{1260}x^8, \\ & u_5(x) = \frac{62}{14175}x^{10}, \\ & u_6(x) = \frac{691}{467775}x^{12}, \\ & \ldots. \end{aligned} \tag{44}$$

More components in the decomposition series can be computed to enhance the approximation. In view of (44), the solution $u(x)$ in a series form is readily obtained by

$$u(x) = x^2 + \frac{1}{6}x^4 + \frac{2}{45}x^6 + \frac{17}{1260}x^8 + \frac{62}{14175}x^{10} + \frac{691}{467775}x^{12} + \cdots, \tag{45}$$

or equivalently

$$u(x) = -2\left(-\frac{1}{2}x^2 - \frac{1}{12}x^4 - \frac{1}{45}x^6 - \frac{17}{2520}x^8 - \frac{31}{14175}x^{10} - \frac{691}{935550}x^{12} + \cdots\right). \tag{46}$$

The exact solution is given by

$$u(x) = -2\ln(\cos(x)). \tag{47}$$

It is interesting to point out that the solution (47) is bounded in the domain $0 \leqslant x \leqslant 1$.

6. Discussions

The goal to obtain exact solutions for Bratu-type problems by using Adomian decomposition method has been achieved. The method has been applied directly without using linearization or any restrictive assumptions. The solution for the first Bratu-type problem blows up at the middle of the domain and remains bounded elsewhere. However, the solutions for the second Bratu-type model and the initial value problem is bounded in the whole domain. The computational size is reasonable compared to other techniques. This confirms the fact that was emphasized by many authors that Adomian decomposition method is effective and reliable.

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